Jason St George

Principal Systems Architect — AI, Autonomy & High-Constraint Infrastructure

jasonstgeorge.com | GitHub | LinkedIn | jason@jasonstgeorge.com | (585) 867–1087

U.S. Citizen

Summary

Systems architect, technical founder, and ML/software engineer with 7+ years building intelligent and distributed systems under difficult operational constraints. I specialize in ambiguous problems where architecture matters: real-time sensor AI, air-gapped ML infrastructure, adversarial distributed systems, agent and retrieval infrastructure, and 0→1 product platforms. I work from system definition through critical implementation, deployment, and operational handoff.

Signature Outcomes

- Led a DSP+ML real-time detection pipeline achieving **sub-50 ms Jetson inference**, sustained end-to-end on embedded hardware under operational noise.
- Built distributed LLM fine-tuning and serving for air-gapped, restricted environments across a **multi-GPU cluster**; reduced fine-tuning costs approximately **65%** through quantized adapter-based methods.
- Built and led production of a **ciphertext-only adversarial storage network** with ongoing integrity proofs, anti-precompute challenges, and tiered incentives, operated in production under continuous verification.
- Co-founded Alchemical AI and built **TurnkeyHQ**, a multi-tenant vertical AI operating system, owning product architecture, application and data layers, integrations, release infrastructure, tenant deployment, QA, and lifecycle management.

Work Experience

Co-Founder & Principal Engineer 2024–Present

Alchemical AI LLC | TurnkeyHQ case study | alchemicalai.com

- Conceived and built TurnkeyHQ for real-estate professionals, carrying the product from market thesis and domain model through production deployment, customer configuration, maintenance, and commercial iteration.
- Owned the complete technical surface: multi-tenant application architecture, agent workflows, data model, auth, third-party integrations, observability, migrations, infrastructure, release engineering, QA, and incident response.
- Developed repeatable tenant onboarding and deployment across a 14-service topology and nine deployable images; created controls for build provenance, environment validation, smoke testing, rollback readiness, and post-deployment verification.
- Managed roadmap tradeoffs and the full product lifecycle while serving as the engineering, platform, release, and QA function.

Senior Machine Learning Software Engineer July 2020–Present

Black River Systems Co., Utica, NY

- **Owned** distributed LLM fine-tuning capability in an air-gapped, restricted environment, from data pipelines and training orchestration through deployment and operator workflows.
- Shipped online learning for fleets of edge devices with bandwidth-aware synchronization; reduced data egress **60%** while maintaining model quality.
- Delivered Jetson-deployed real-time signal classification combining DSP feature extraction, DeepCNN inference, and end-to-end latency measurement for real-time operator workflows.

Work Experience (continued)

Data Scientist I June 2019–May 2020
Northeast Information Discovery, Inc., Canastota, NY

- Improved RF modulation classification accuracy 12pp using attention-based spectral encoders; built APIs and deployed models into streaming production environments with R, Python, and C++.

Selected Systems & Research

Agentic Data | jasonstgeorge.com/agentic-data 2025

- Architected an enterprise context graph and retrieval planner: 27 endpoints, 21 Postgres tables, 35 edge types, five retrieval strategies, multi-hop expansion, institutional connectors, and 315 tests.

SwarmOS | jasonstgeorge.com/swarmos 2025

- Built an auditable multi-agent research substrate: 13-package TypeScript monorepo, 30+ authenticated APIs, event-stream coordination, content-addressed artifacts, structured claim graphs, verification grades, and budgeted execution.

AfterFiat — Sole Author, v3.4 | Research profile 2024–Present

- Wrote and maintains a **checksummed, versioned monetary research thesis** testing whether Privacy, Proofs, and Compute can support store-of-value premium under sustained repression: eleven premises, a nine-link chain, eighteen red lines, and four telemetry families.
- Separates utility from moneyness, protocol value capture from market price, and duration-neutral base assets from duration-bearing infrastructure credit; publishes section-level citations and formal release notes.

Published Research Google Scholar

Music Style Transformer — St. George, J., Bischof (2019). Bayesian-factorized raw-audio transcription, composition, and synthesis; transformer-based generation. *ICAI '19*, pp. 22–33.

Sonification of Black Hole Merger Data — St. George, J., Bischof, H.P., Kim, S.Y. (2018). Velocity-modulated synthesis and HRTF spatialization of CCRG simulations. *MSV '18*, pp. 3–9.

Technical Capabilities

- **Languages / AI:** Python, TypeScript/JavaScript, Rust, SQL, C++; PyTorch, LoRA/QLoRA/PEFT, edge inference, DSP+ML, retrieval planning, agent orchestration, evaluation
- **Platform / Data:** FastAPI, PostgreSQL/pgvector, Neo4j, OpenSearch, Redis Streams, MinIO/S3, Docker, Kubernetes, Slurm, JWT/RBAC, CI/CD, observability
- **Optimization / Protocols:** OR-Tools CBC, MILP, Pedersen commitments, Merkle proofs, challenge-response verification, incentive design, threat modeling

Education

M.S., Computer Science — Rochester Institute of Technology May 2019
B.M., Performance; Music Theory Minor — University of North Texas May 2013